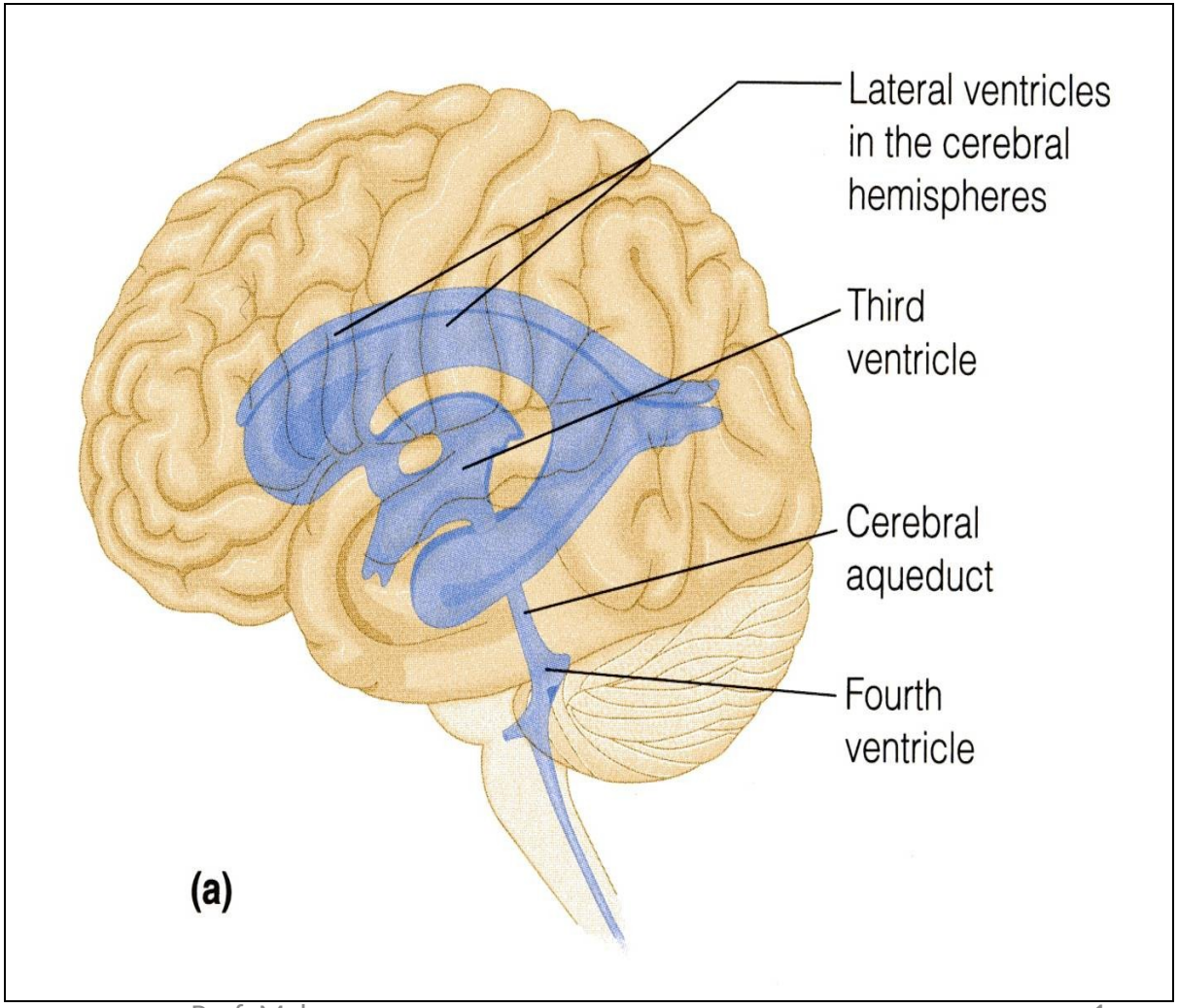
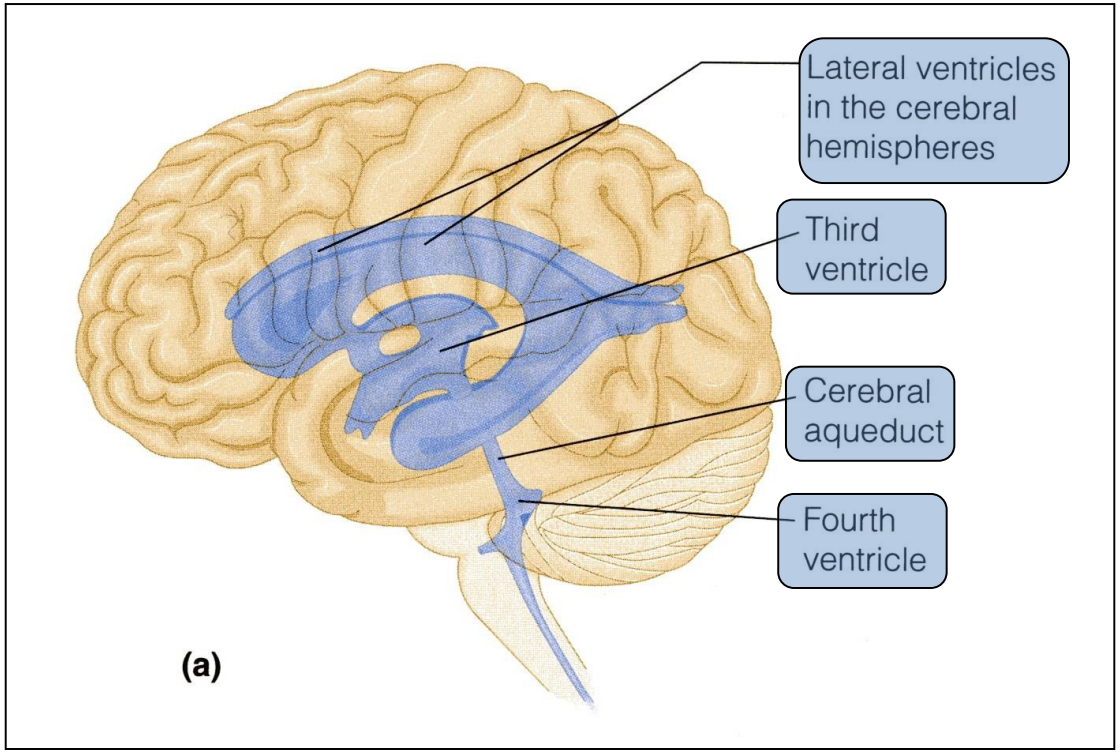
Lateral Ventricle



By
Prof. Saeed Abuel Makarem

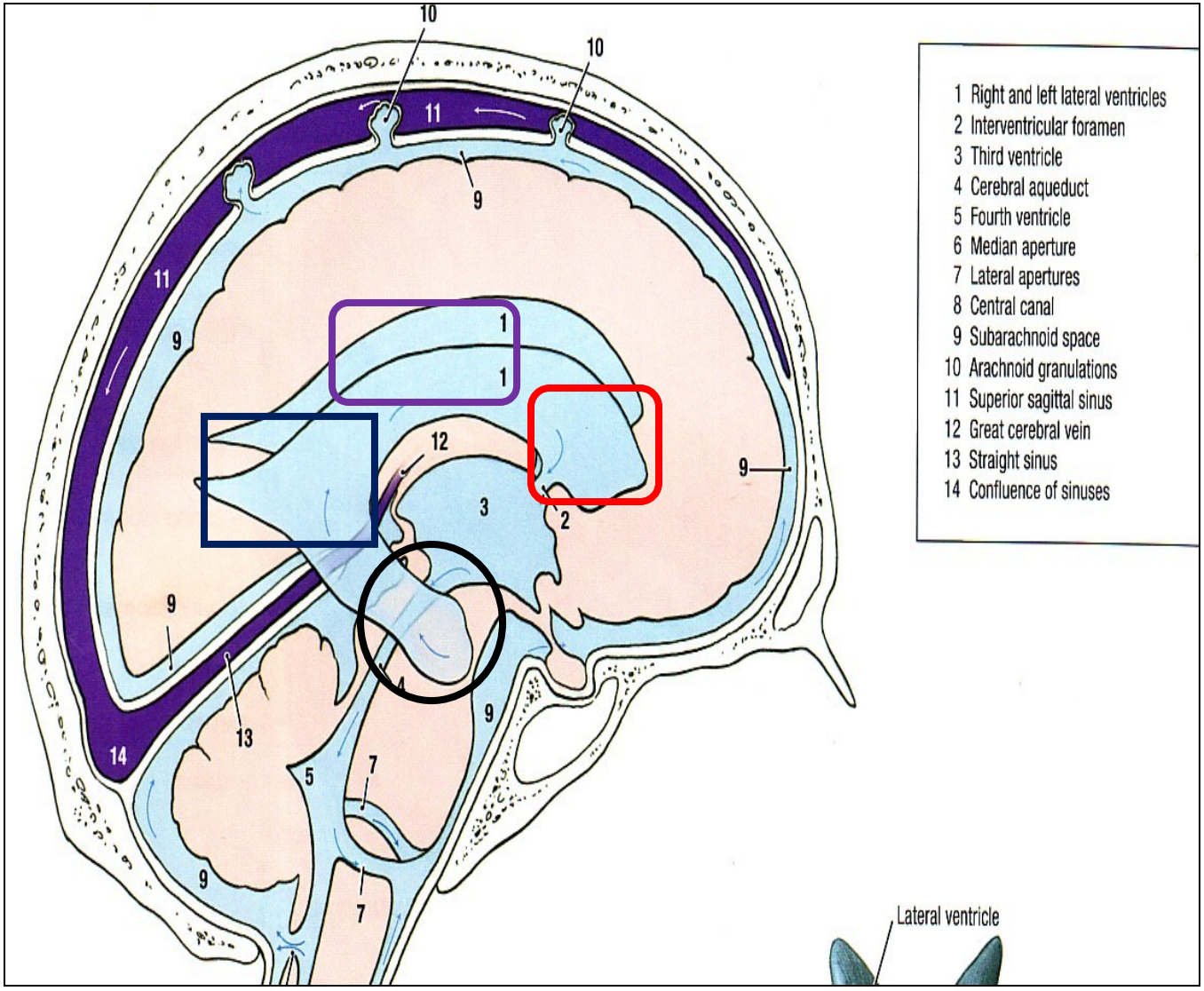
BRAIN VENTRICLES



- The brain is bathed by the **cerebrospinal fluid** (CSF)
- Inside the brain, there are spaces **(ventricles)** filled with CSF
- There are 4 ventricles
  - 2 **lateral ventricles** are in the brain hemispheres
  - **3rd ventricle** is in the diencephalon
  - **4th ventricle** is between the pons, open medulla and the cerebellum
  - The 3rd & the 4th ventricles are connected by the **cerebral aqueduct**

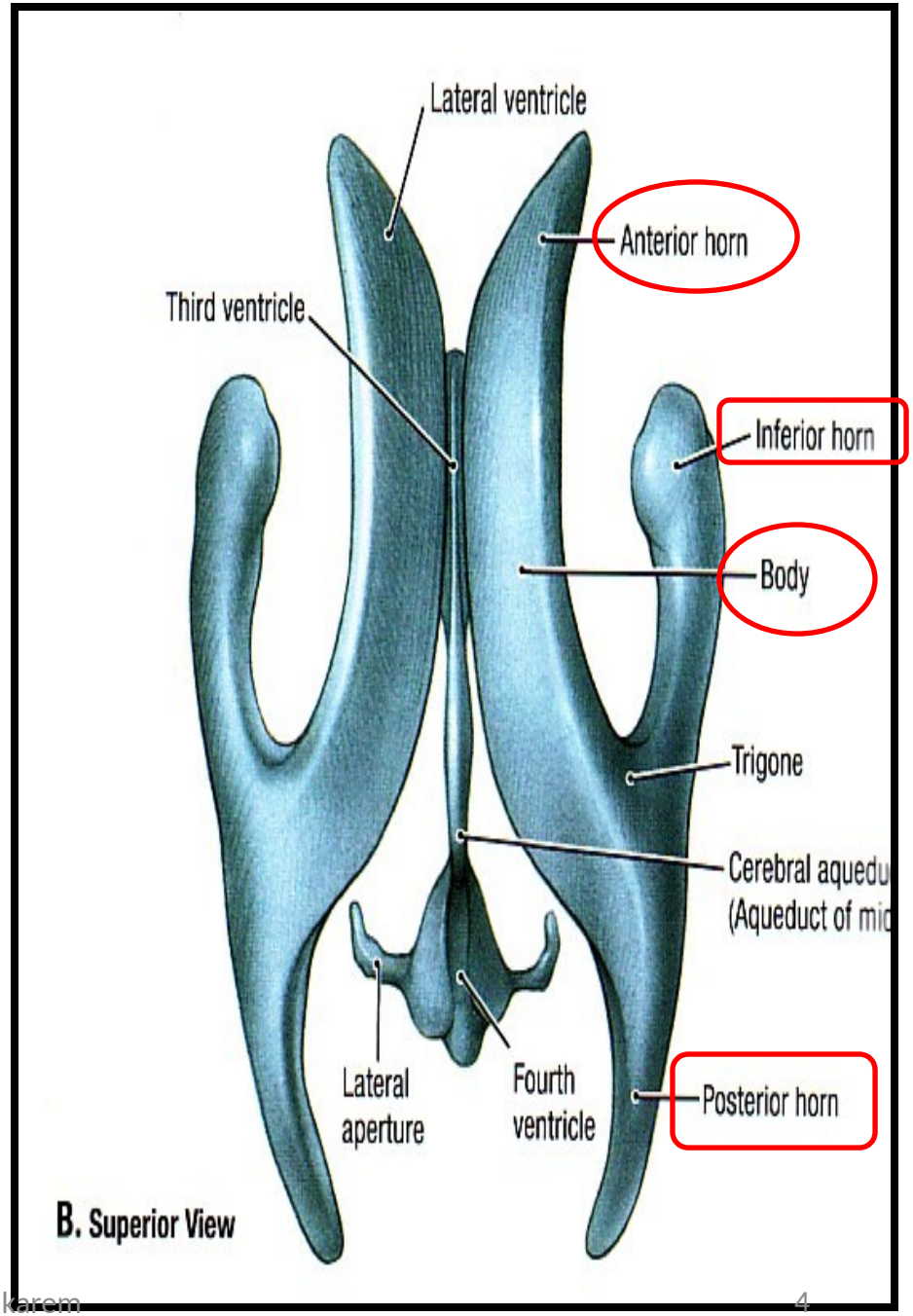
Lateral Ventricle

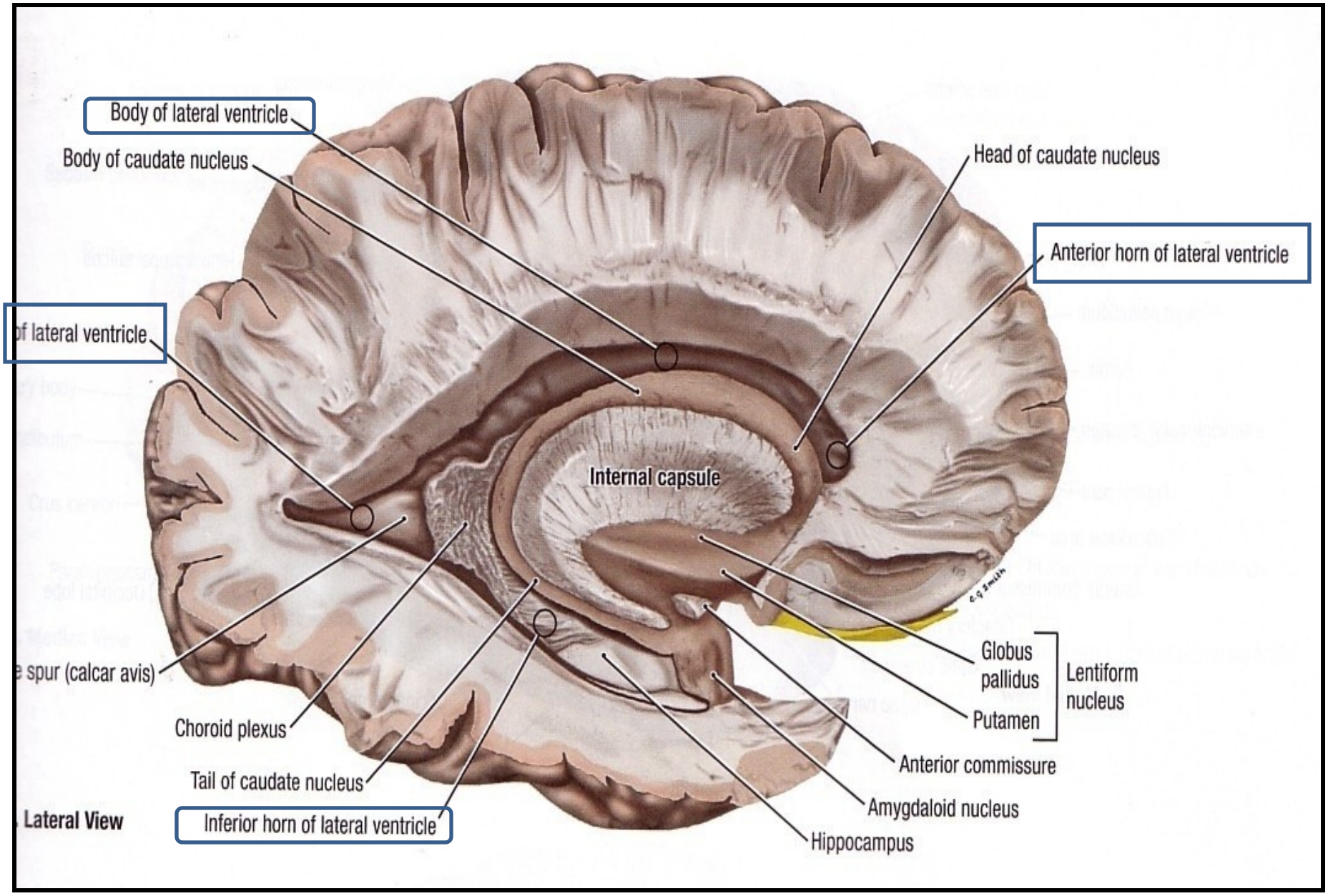
- **Definition :**
- It is the cavity of the cerebral hemisphere.
- It is C-shaped.
- It has 3 horns **&** central part.
- **Anterior Horn**: in the frontal lobe.
- **Posterior horn**: in the occipital lobe
- **Inferior horn**: in temporal lobe.
- **Central part or body**: in the parietal lobe.



Lateral Ventricle

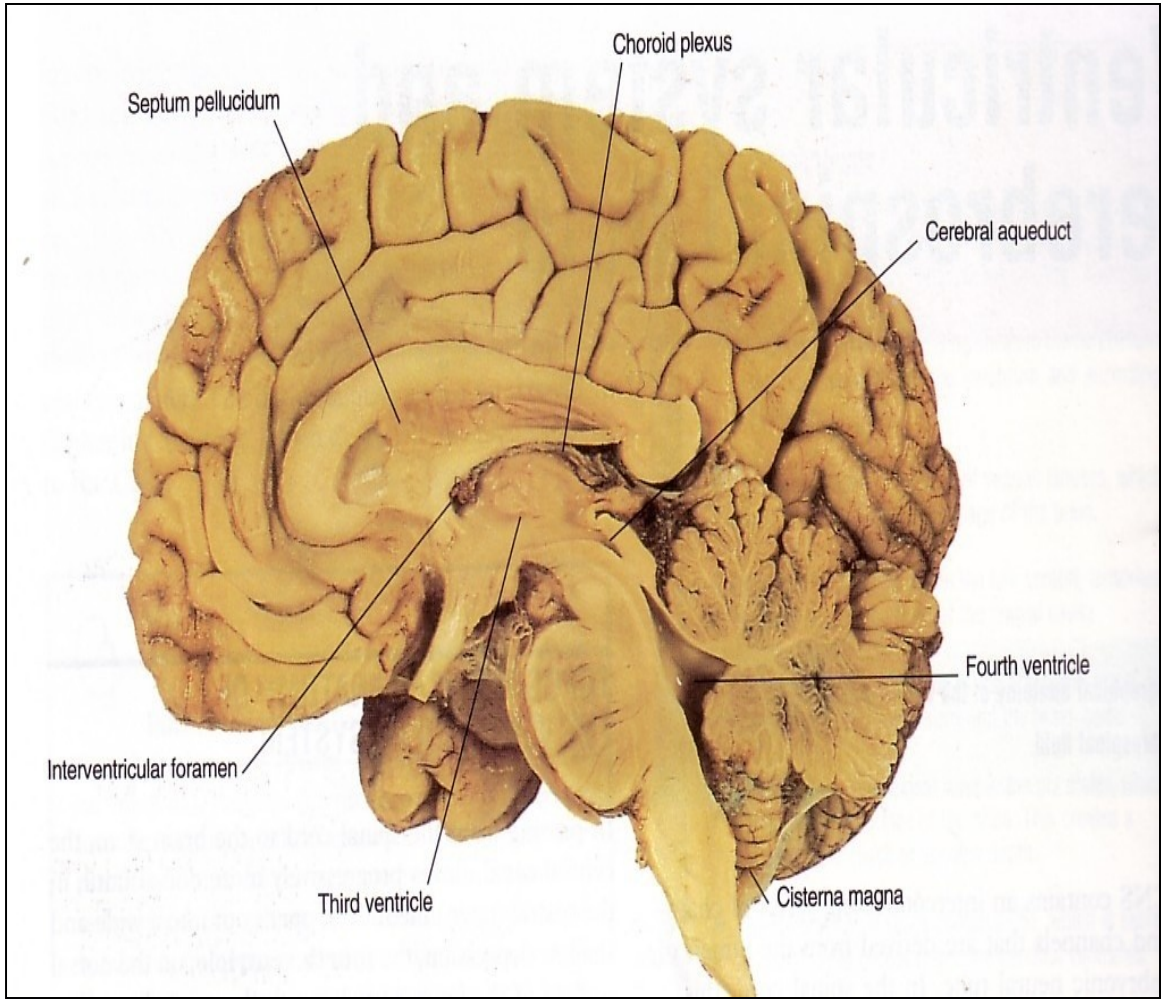
- Superior view of the ventricular system.
  - **Lateral ventricle**
- Anterior horn in the frontal lobe.
- Posterior horn in the occipital lobe.
- Inferior horn in the temporal lobe.
- Body: In the parietal lobe.
- The inferior and posterior horns are connected in the trigon.

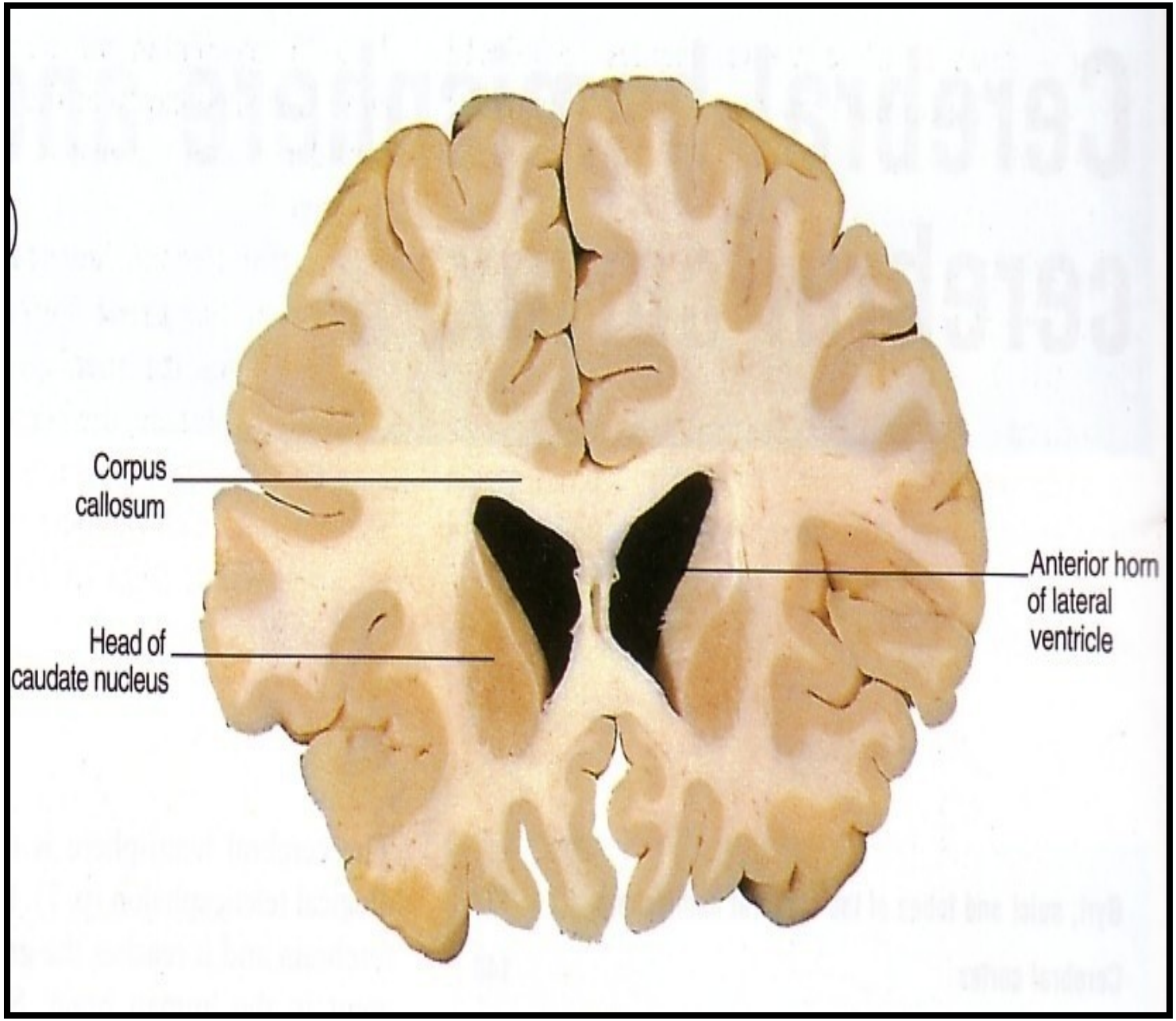



Body of lateral ventricle
Body of caudate nucleus
Head of caudate nucleus
Anterior horn of lateral ventricle
of lateral ventricle
Internal capsule
e spur (calcar avis)
Globus pallidus
Lentiform nucleus
Putamen
Choroid plexus
Anterior commissure
Tail of caudate nucleus
Amygdaloid nucleus
Inferior horn of lateral ventricle
Hippocampus
Lateral View

Anterior Horn

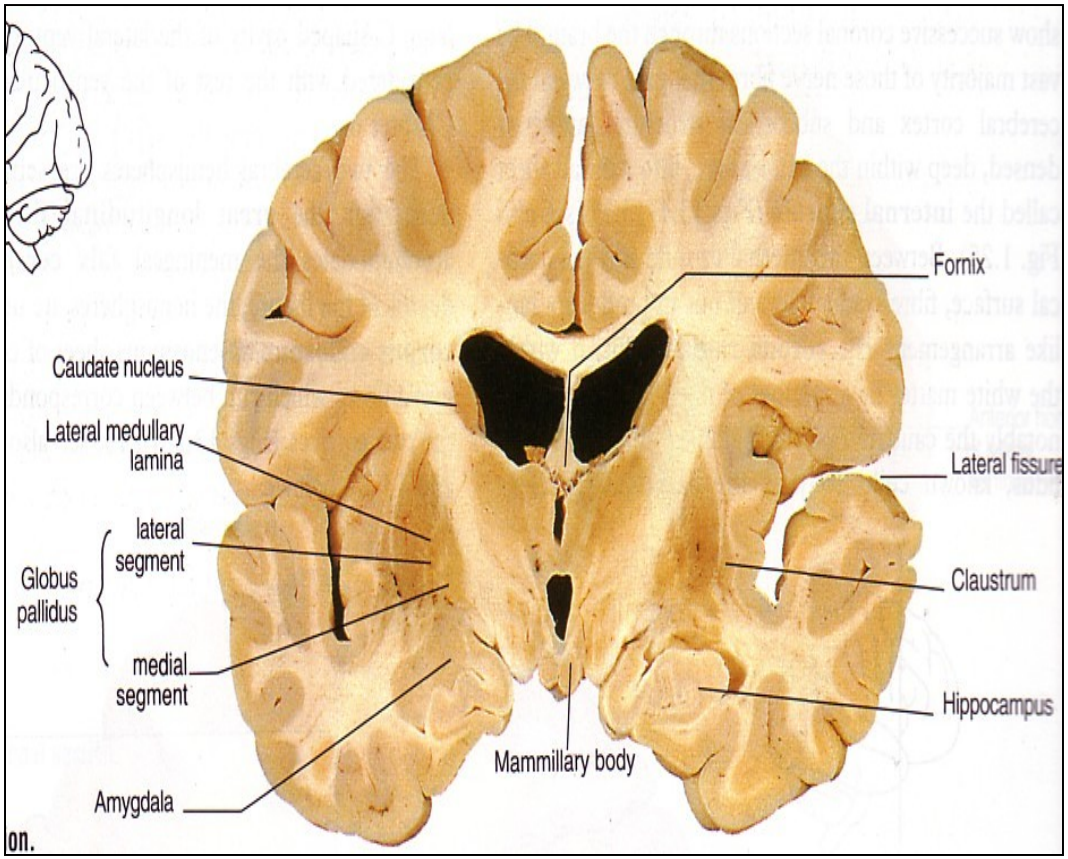
- In the frontal lobe.
- Roof:
- **Corpus callosum** (trunk)
- Floor:
- **Corpus callosum** (Rostrum)
- Anterior:
- **Corpus callosum** (Genu)
- Medially:
- Septum pellucidum.
- Laterally:
- ***Head*** of ***Caudate nucleus***.




Corpus callosum
Head of caudate nucleus
Anterior horn of lateral ventricle

Body or Central part

- Lies in the parietal lobe.
- **Roof:**
- **Corpus callosum** (Trunk).
- **Floor:**
- Sloping, From lateral to medial it is formed by:
- Body of caudate nucleus,
- Upper surface of thalamus
- Choroid plexus,
- Body of fornix.
- **Medial wall:**
- Septum pellucidum.
- **Lateral wall:**
- *narrow area at the* meeting of roof & floor.

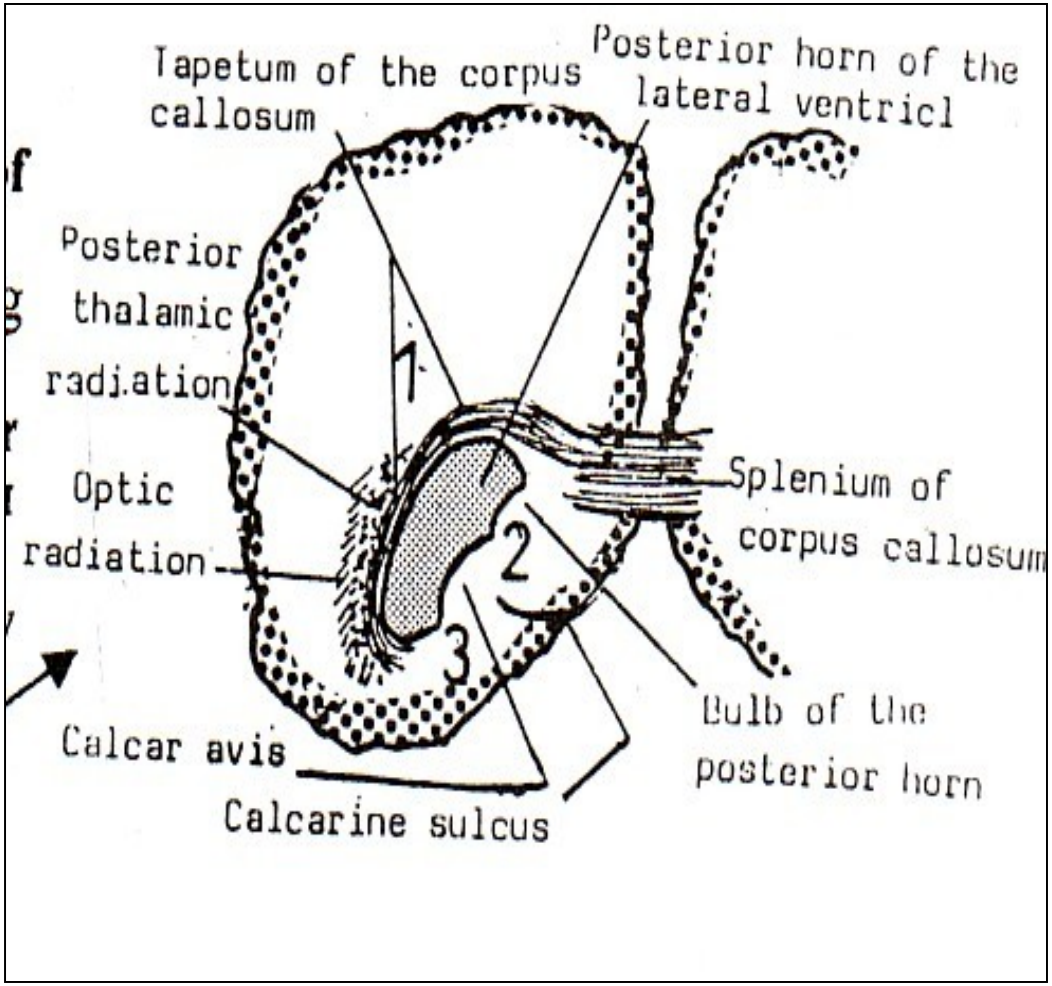


Posterior Horn

- In the occipital lobe.
- Roof, lateral wall, and floor:
- Are formed by the Tapetum of the corpus callosum.
- Medially:
- There are 2 elevations:

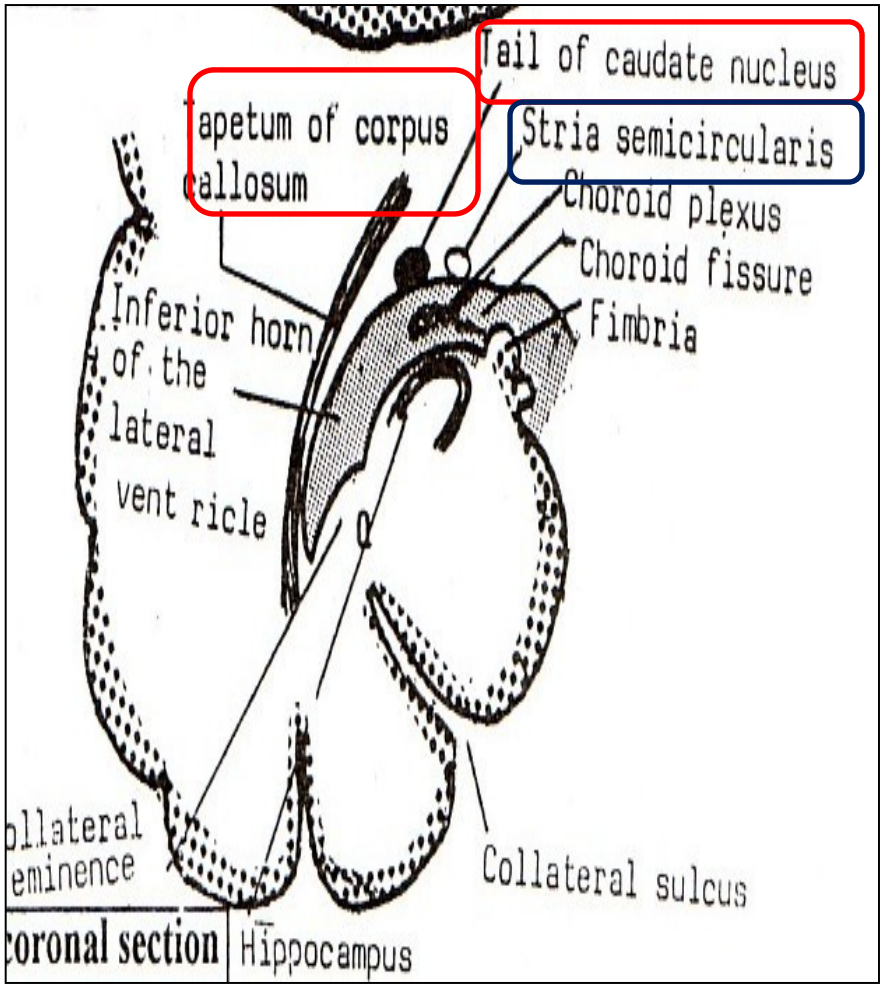
Bulb of posterior horn (formed by forceps major-**2**-).

Calcar avis: produced by calcarine sulcus-3-.

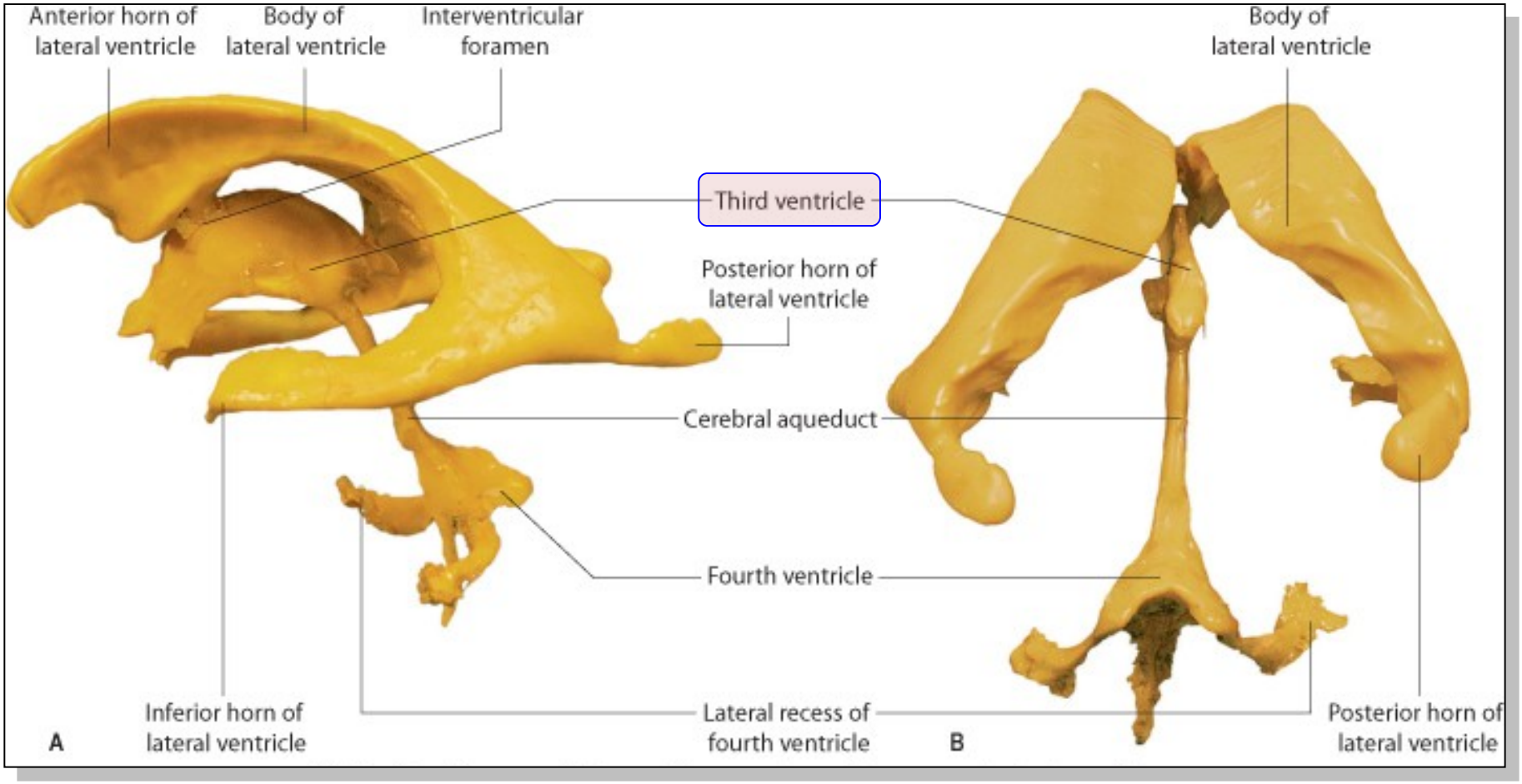


Inferior Horn

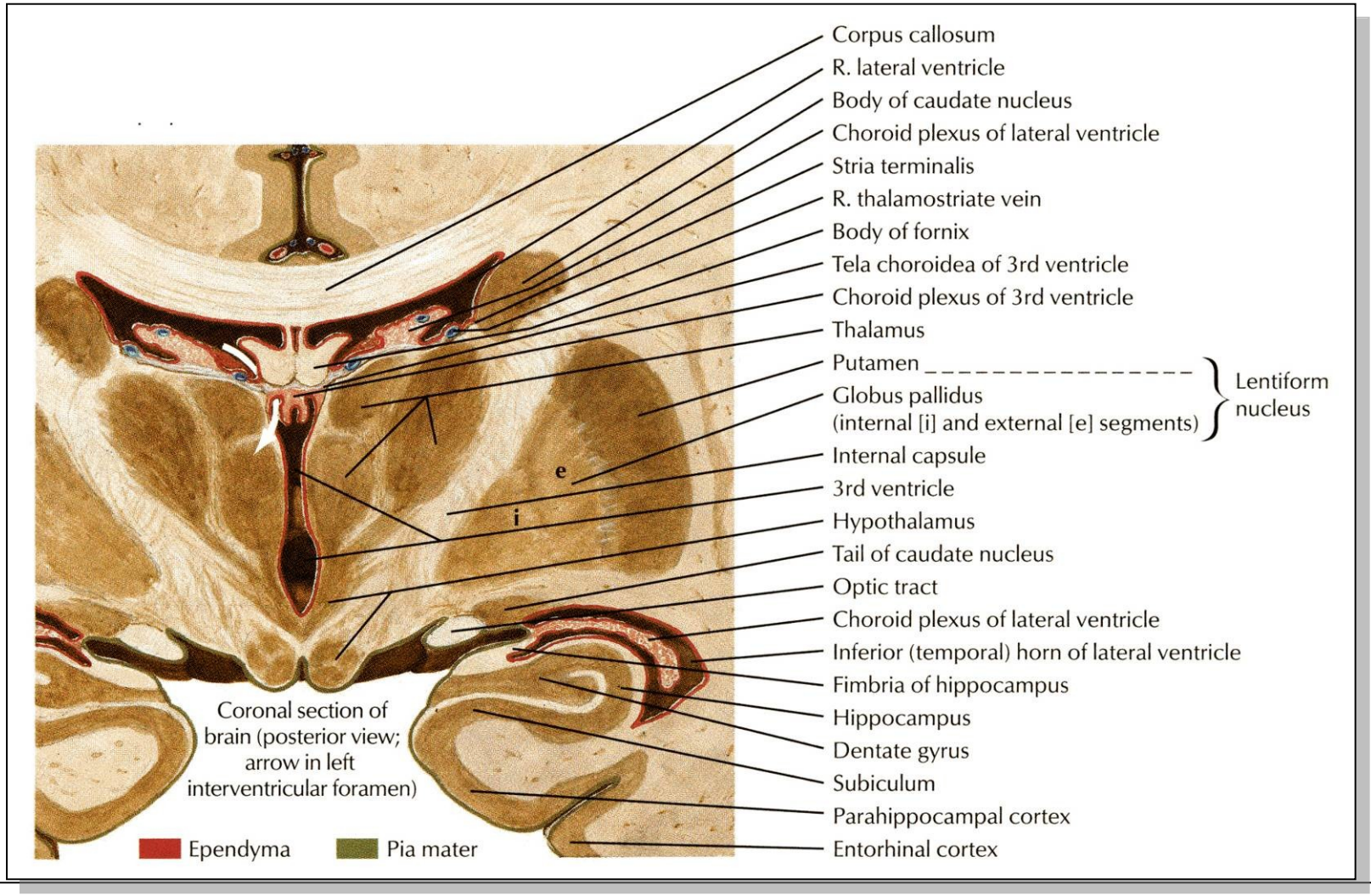
- It lies in the temporal lobe.
- **Roof:**
- Tapetum,
- Tail of caudate nucleus,
- Amygdaloid nucleus
- Stria terminalis.
- **Floor:**
- Hippocampus,
- Fimbria of hippocampus & Collateral eminence.
- **Lateral wall:**
- Tapetum of the corpus callosum.



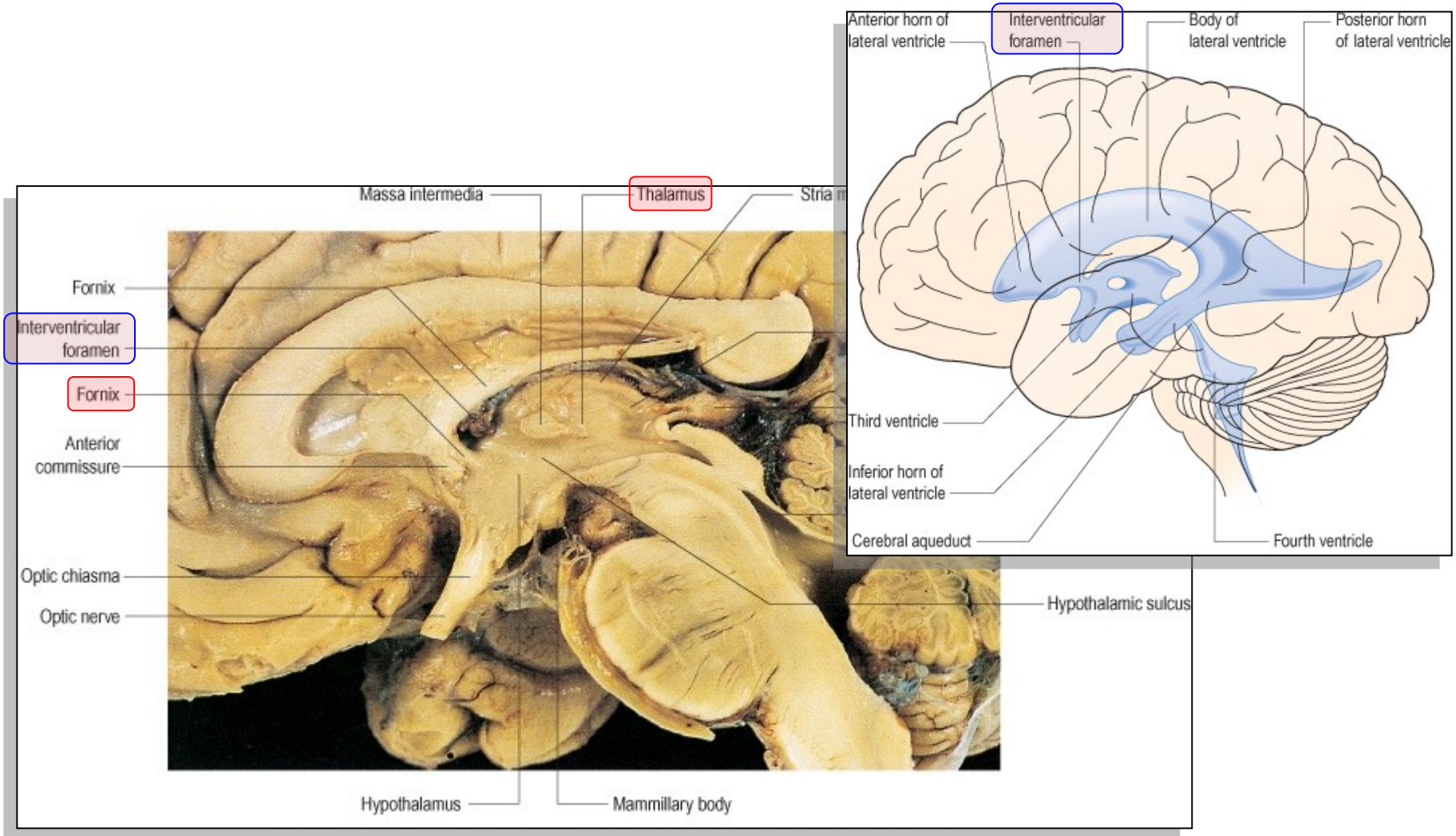
3rd VENTRICLE



Third ventricle is a narrow slit-like cavity whose lateral walls are formed by the thalamus and hypothalamus on either side. At the rostral margin of the midbrain, the cerebral aqueduct opens into the **third ventricle.**

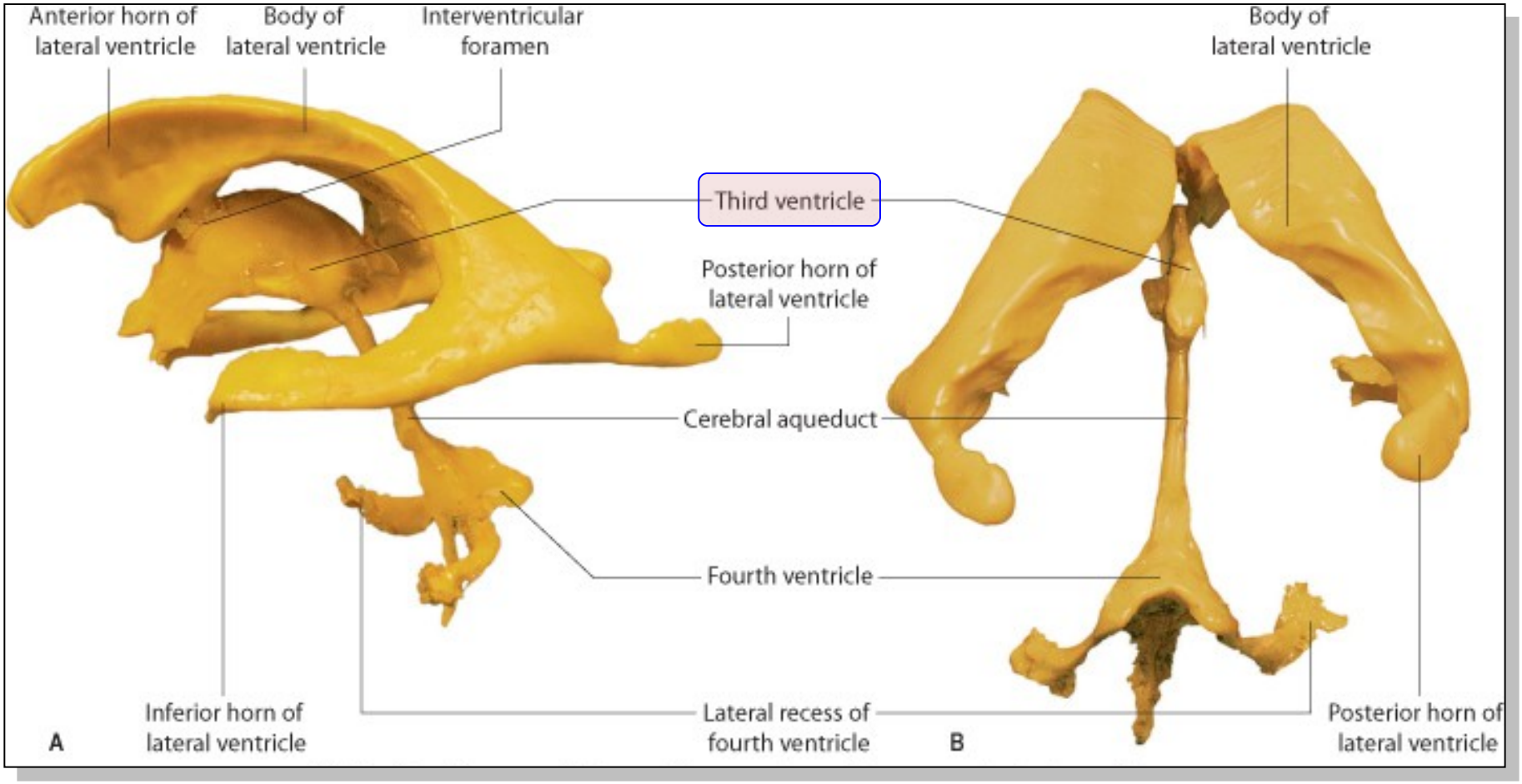


The roof of the ventricle is formed by pia-ependyma, which spans between the two striae medullaris thalami, situated along the dorsomedial border of the thalamus.

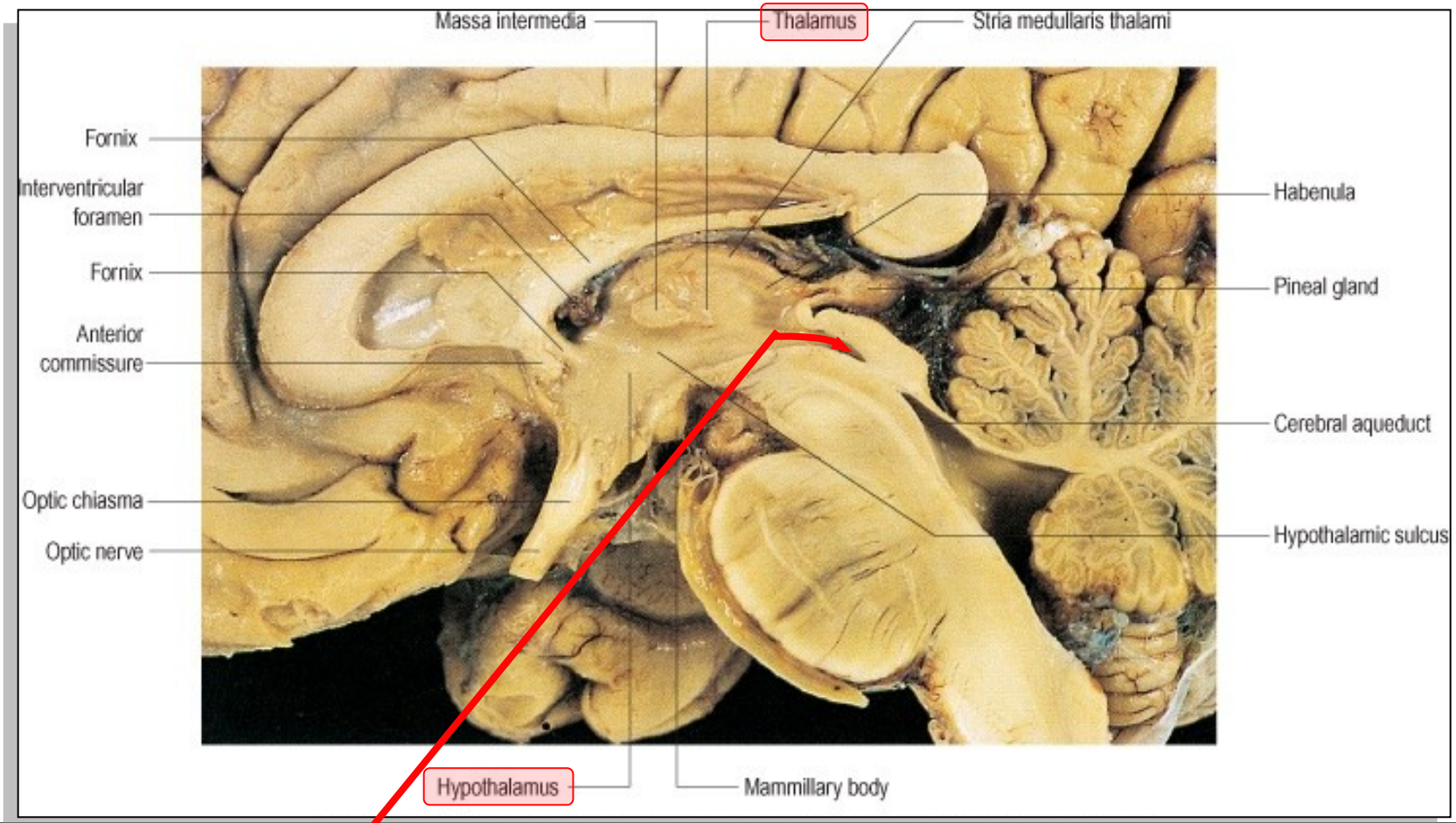


In the rostral part of the third ventricle lies an aperture, the **interventricular foramen** or **foramen of Monro**, which is located *between the column of the fornix and the anterior pole of the thalamus.*

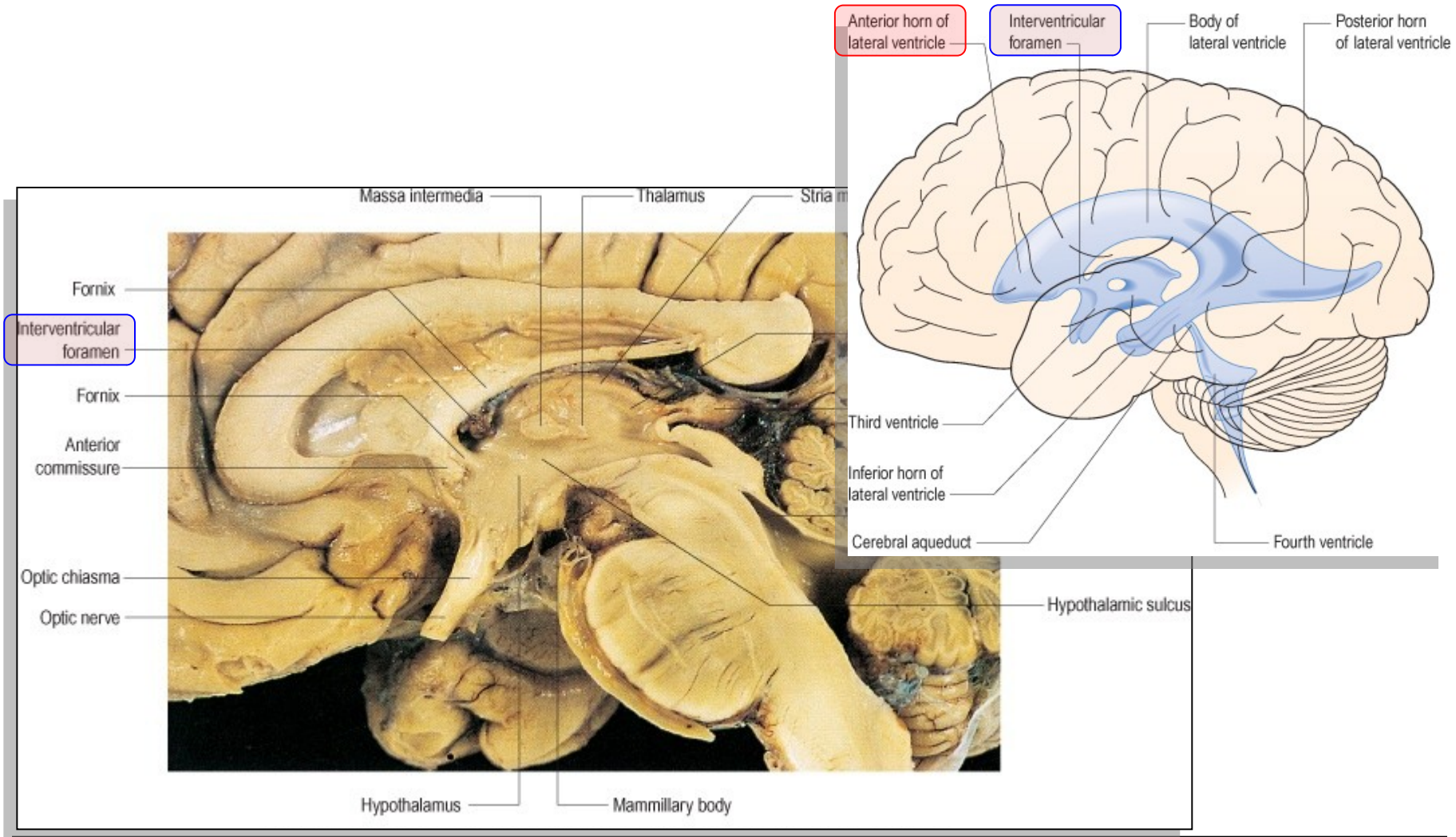
TOPOGRAPHICAL ANATOMY



The third ventricle is a midline, slit-like cavity.



- The **lateral walls** of the 3rd ventricle is formed of the *thalamus and hypothalamus.*
- **Caudally,** the third ventricle becomes *continuous with the cerebral aqueduct.*



The **interventricular foramen** *provides communication*, on either side, *with the* extensive *lateral ventricle* located within the cerebral hemisphere.